

Applying Cluster Analysis in R for Business Segmentation

Data Analysis for Decision-Making (SS 2026) | Prof. Dr. Stephan Huber

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Introduction

Imagine this scenario... A retail company has 10,000 customers.

Every customer has different:

- Income levels
- Spending habits
- Buying preferences

Should the company send the same promotion to everyone?

No.

Businesses need to identify groups of similar customers so they can create targeted marketing strategies.

What is Cluster Analysis?

Cluster Analysis is an unsupervised machine learning technique that groups similar observations together based on their characteristics.

Its objective is to: -

- Maximize similarity within each group
- Minimize similarity between different groups

Unlike classification, there are no predefined labels.

Why is Business Segmentation Important?

Business segmentation helps organizations:

- Understand customer behavior
- Personalize marketing campaigns
- Improve customer satisfaction
- Increase sales

Instead of treating every customer equally, businesses can focus on the needs of each segment.

Applications of Cluster Analysis

Cluster analysis is widely used in:

- Customer segmentation
- Product recommendation systems
- Fraud detection
- Credit risk analysis
- Market research
- Healthcare analytics
- Social media analysis

Workflow of Cluster Analysis

The complete process follows these steps:

1. Collect data
2. Prepare and clean data
3. Scale variables
4. Choose the optimal number of clusters
5. Apply K-Means algorithm
6. Interpret business insights

Data Preparation

Before clustering:

- Remove missing values
- Select relevant variables
- Standardize the data

Scaling is the important because variables may have different ranges.

For example:

Income = 20-150

Spending score = 1-100

Without scaling, income would dominate the clustering process.

Choosing the Number of Clusters

One of the biggest question is:

How many clusters should we create?

- The most common approach is the Elbow Method.
- It calculates the Within-cluster Sum of Squares (WSS) for different values of K.
- The point where improvement starts slowing down is called Elbow.
- That value is selected as the optimal number of clusters.

How K-Means Works

The K-Means algorithm follows four simple steps:

1. Select K initial centroids
2. Assign each data point to its nearest centroid
3. Recalculate centroid positions
4. Repeat until the assignments stop changing

The final result is a set of well-defined customer groups.

Interpreting Customers Segments

Example segmentation:

Cluster 1 High Income + High Spending -> Premium Customers

Cluster 2 High Income + Low Spending -> Up-selling Opportunities

Cluster 3 Low Income + High Spending -> Promotion-Sensitive Customers

Cluster 4 Low Income + Low Spending -> Budget Customers

Each group requires a different marketing strategy.

Business Benefits

Using cluster analysis helps businesses:

- Target the right customers
- Improve marketing effectiveness
- Increase customers retention
- Allocate resources efficiently
- Discover hidden patterns in customer behavior

Ultimately, it leads to better business decisions.

Advantages

- Easy to understand
- Simple to implement
- Scaleable to large data-sets
- Produces meaningful customers segments
- Supports personalized marketing

Limitations

- Selecting the correct number of clusters can be difficult
- Sensitive to outliers
- Requires proper data scaling
- Different initial centroids may produce different results.

Therefore, validation is always important.

Conclusion

Cluster analysis transforms raw customer data into valuable business insights.

By identifying customer segments, organizations can make smarter decisions, optimize marketing campaigns, and improve customer relationships.

Thank you

Questions?